

TRUST-BC: A Multicohort Benchmark for Leakage-Free and Calibrated Breast Cancer Prediction

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ABSTRACT

Background: Models using machine learning for breast cancer prediction are often reported with accuracies over 97%, but such reports often mask preprocessing leakage, uncalibrated probabilities, and misclassification of outcomes that impede clinical translation, as recently illustrated by a reanalysis in which correcting leakage reduced a reported AUROC from 0.998 to 0.715.

Methods: We present **TRUST-BC**, an open-source multicohort benchmark and reusable auditing template for three structurally different breast-cancer data types—cytopathological (WBCD, $n = 569$), metabolic (Coimbra, $n = 116$), and epidemiological registry (SEER, $N = 3,270$)—that jointly implements leakage-free nested cross-validation, corrected equivalence testing, distribution-free conformal uncertainty, multidimensional calibration, and censoring-aware survival modeling in a single pipeline.

Results: The leakage-free Unified pipeline matched or approached leaky-baseline performance on WBCD (accuracy 97.08%) and Coimbra (71.35%), and reached AUROC 0.732 on SEER, but formal statistical equivalence with the leaky baseline held only for WBCD—revealing that leakage-driven optimism is real and cohort-dependent rather than a uniformly negligible artifact. Conformal prediction classified the majority of SEER predictions (51.38%) as clinically ambiguous compared to 0.90% for WBCD, directly reflecting the inherent discriminability of each cohort and revealing the points where automated point predictions are least reliable. This signal was confirmed by an independent Weibull survival branch (C -index 0.722, log-rank $p = 1.02 \times 10^{-9}$). The framework’s split-conformal layer further operationalizes predictive uncertainty as a two-stage clinical triage engine, designed to route high-confidence predictions to standard pathways while flagging uncertain cases for expert review—a capability absent or unassessed in the comparator studies summarized later in this manuscript.

Conclusion: TRUST-BC demonstrates that leakage risk, calibration reliability, and prediction uncertainty need to be assessed per cohort and cannot be assumed homogeneous, and provides a publicly available template, compliant with TRIPOD+AI and PROBAST+AI guidelines, that other prognostic ML studies can directly adopt or benchmark against.

Keywords: Breast Cancer, Machine Learning, Data Leakage, Conformal Prediction, Probability Calibration.

1 Introduction

1.1 Clinical Motivation and Computational Pitfalls in Oncology Modeling

In 2022, breast cancer was the most commonly diagnosed cancer and the leading cause of cancer death among women worldwide, with an estimated 2.3 million new cases and approximately 670,000 deaths [1]. Accurate early diagnosis and long-term prognostic risk assessment are essential for the design of standard treatments, including surgical resection, systemic adjuvant chemotherapy, and targeted endocrine therapies.

Over the past decade, high discriminative performance has been achieved by supervised machine learning (ML) models (support vector machines, tree ensembles, and gradient boosting architectures [2, 3]) on retrospective tabular oncology benchmarks.

Despite the promising benchmark metrics of these algorithms, their translation into clinical practice is still challenging. Widespread methodological vulnerabilities often impair reproducibility and translational validity, as evidenced throughout the biomedical machine-learning literature [4]. Four structural vulnerabilities are particularly common in clinical oncology workflows:

- **Parametric and Covariance Data Leakage:** Preprocessing transforms like standard scaling and principal component analysis (PCA) are often fit on the entire dataset before splitting the data. Even if sample IDs do not overlap between splits, global distributional statistics leak across folds, leading to unearned variance reductions and overly optimistic performance estimates.
- **Probability Miscalibration:** The standard optimization objectives focus on discriminative metrics, e.g., accuracy and AUROC, but neglect probability calibration [5, 6]. Thus, the predicted probabilities often do not match the true empirical rates of events.
- **Censoring Misclassification in Registries:** In the conversion of time-to-event cohorts to binary classification targets, patients who are right-censored and drop out of monitoring before the study endpoint are often incorrectly labeled as non-events, causing systematic label noise.
- **Lack of Clinical Uncertainty Quantification:** Classifiers yield deterministic point probabilities without distribution-free coverage guarantees, leaving clinical practitioners without safety filters for borderline or atypical presentations [7].

1.2 Related Work and Methodological Deficits in Current Literature

Machine learning algorithms have been applied to tabular breast cancer cohorts in many studies with varying methodological rigor. On the morphological WBCD benchmark, reported accuracies often exceed 97%, but unsupervised feature transforms seldom occur within cross-validation loops of pipelines, with the potential for covariance leakage. Published models evaluating multi-biomarker panels on serum metabolic profiles (Coimbra cohort) have often relied on unnested or uncalibrated splits [10]. Studies of large population-scale cancer registries, such as SEER, frequently transform follow-up records into 5-year survival targets, without accounting for patients censored prior to 60 months, resulting in distorted ground-truth outcome distributions.

Recent prognostic studies, including the comparative re-analysis benchmark by Tadj et al. (2026) [15], have shown an increasing interest in systematic evaluation. However, many existing studies still rely on heuristic shortcuts. First, a standard approach is to compare cross-validation fold scores using paired t -tests without correcting for the substantial positive correlation induced by overlapping training sets, which can inflate the Type I error rate [21]. For single-split comparisons on a fixed test set, DeLong’s test provides a valid nonparametric comparison of correlated AUROCs [26]; however, it does not resolve the fold-correlation problem when applied naively across repeated cross-validation folds. Second, models often produce point probabilities with no uncertainty intervals or triage mechanisms. Third, sensitivity to sample scale and feature-subspace dimensionality is largely unquantified. Table 5 (Section 4) presents a quantitative, multi-domain comparison of existing literature and the safeguards established in this work.

This disconnect is not unique to oncology. A recent systematic review and meta-analysis of 21 included cardiovascular ML studies using electronic health records reported substantial heterogeneity ($I^2 > 99\%$) and potential publication bias across studies, concluding that despite enhanced calibration in some ML models, methodological concerns currently limit clinical applicability [16]. Cai et al. [17] separately identified recurrent cardiovascular ML pitfalls involving data quality, small samples and low event rates, overfitting, inadequate evaluation criteria, and limited generalizability. Among the remedies discussed, the authors explicitly recommended appropriate validation measures and listed the separation of training, tuning, and testing data as an approach to reducing overfitting. These parallel observations in a distinct clinical domain indicate that the vulnerabilities addressed by TRUST-BC are not particularities of the breast-cancer literature, but part of a systemic pattern across clinical ML, supporting the need for a reusable, domain-agnostic auditing template rather than a one-off correction.

1.3 Objectives and Key Contributions

Unlike prior single-cohort leakage audits [15] that addressed pre-split oversampling contamination and temporal misalignment within a single epidemiological registry, this work presents **TRUST-BC**—a

comprehensive, open-source auditing and modeling template that concurrently addresses parametric leakage, censoring-driven label noise, probability miscalibration, and the lack of distribution-agnostic uncertainty quantification across three different archetypes of data (cytopathological, metabolic, and epidemiological registry).

This study makes six key methodological contributions:

1. **Parametric Isolation by Construction:** Feature standardization, dimensional reduction, and hyperparameter tuning are strictly encapsulated into the inner cross-validation folds through the use of scikit-learn pipeline architectures [8] without parametric and covariance leakage by construction.
2. **Equivalence and Superiority Hypothesis Testing:** We evaluate model differences using the Nadeau–Bengio corrected resampled t -test [21] with Holm–Bonferroni error control [22] combined with Two One-Sided Tests (TOST) [23, 24] with pre-specified equivalence margins ($\Delta = 0.02$ for accuracy, $\Delta = 0.01$ for AUROC) to determine whether leakage elimination maintains the true predictive signal.
3. **Finite-Sample Uncertainty Quantification:** We employ inductive split conformal prediction [27, 28] to generate prediction sets with finite-sample coverage guarantees ($1 - \alpha = 95\%$) and flag ambiguous cases for expert review.
4. **Full Calibration Assessment and Decision Analysis:** We assess predictions using Brier scores, Expected Calibration Error (ECE) [29], calibration slope and intercept, and Decision Curve Analysis (DCA) [32] for clinical net benefit.
5. **Censoring-Aware Registry Filtering and Survival Modeling:** We remove right-censored label corruption for 5-year classification in the SEER population cohort and use a penalized Weibull Accelerated Failure Time (AFT) model with Kaplan–Meier risk stratification for continuous time-to-event prediction.
6. **Methodological Reporting Compliance:** The study was designed and reported in line with the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis + AI (TRIPOD+AI) [34] and the updated PROBAST+AI quality, risk-of-bias, and applicability assessment tool [35].

We intentionally bundle these six components into one reusable template rather than treating them as separate studies, because in practice these vulnerabilities co-occur and interact: a model tuned to eliminate leakage, but not also audited for calibration and uncertainty, can create a false sense of methodological rigor. The contribution of TRUST-BC is precisely this joint auditing capability, rather than any individual component in isolation.

Leakage-aware and calibration-aware evaluation frameworks have been proposed for other clinical tabular domains, such as cardiovascular risk prediction and diabetes screening, but these typically address a single vulnerability (leakage, calibration, or uncertainty quantification) in a single cohort. The key novelty of TRUST-BC is the simultaneous, automated treatment of all four vulnerabilities—parametric leakage, censoring-induced label noise, miscalibration, and absence of distribution-free uncertainty—in one auditable pipeline deployed over three structurally different breast cancer data archetypes. To our knowledge, this combination has not been demonstrated before, even outside the literature on oncology.

The principal contribution of TRUST-BC is therefore methodological rather than a claim of superior empirical prediction: it integrates six independently established, but rarely co-implemented, safeguards into a single reusable pipeline that prognostic ML researchers in oncology and other tabular clinical domains can adapt to their own cohorts. In this sense, TRUST-BC is intended primarily as a public auditing framework rather than as a single deployable predictive model—analogous in role, though not in scope, to how TRIPOD+AI and PROBAST+AI support reporting and risk-of-bias assessment rather than functioning as prediction models themselves.

2 Materials and Methods

2.1 Cohort Descriptions and Sample Size Justifications

We evaluated the multicohort framework on three cohorts representing distinct diagnostic and prognostic contexts (Table 1):

Table 1: Cohort Characteristics, Outcome Distributions, and Deterministic Partition Splits.

Cohort	Sample Size (N)	Features (D)	Clinical Endpoint	Class Distribution	Dev Set (85%)	Holdout Lockbox (15%)
WBCD	569	30	FNA Malignancy	357 Benign / 212 Malignant	$n = 483$	$n = 86$
Coimbra	116	9	Serum Cancer Status	52 Control / 64 Cancer	$n = 98$	$n = 18$
SEER (5-Year Mortality)	3,270	16	5-Year Cancer Mortality	2,654 Survived / 616 Deaths	$n = 2,779$	$n = 491$
SEER (Survival Branch)	4,024	12	Time-to-Event Mortality	3,408 Censored / 616 Deaths	$n = 3,219$ (80%)	$n = 805$ (20%)

- Wisconsin Diagnostic Breast Cancer (WBCD):** A classical morphological benchmark of $n = 569$ fine needle aspirate (FNA) samples (357 benign, 212 malignant) collected from the UCI Machine Learning Repository [9, 12]. Features include 30 real-valued morphological measurements of the cell nucleus characteristics, recorded as mean, standard error, and worst values. The sample consists of 212 malignant cases, resulting in an effective events-per-variable (EPV) ratio of 7.1. The subspace projection to $d \leq 15$ dimensions can raise it to ≥ 14.1 effective EPV, while still meeting the sample adequacy criteria.
- Coimbra Biomarker Cohort:** A serum metabolic screening dataset with $n = 116$ subjects (64 breast cancer patients, 52 healthy controls) with 9 routine clinical and metabolic biomarkers: Age, body mass index (BMI), Glucose, Insulin, homeostatic model assessment of insulin resistance (HOMA-IR), Leptin, Adiponectin, Resistin, and monocyte chemoattractant protein-1 (MCP-1) [10, 12]. The EPV is 7.1 with 9 parameters and 64 positive occurrences. This cohort, while small in sample size, is an important archetype for non-invasive screening.
- SEER Registry Research Benchmark:** Based on a curated public research benchmark extraction of the National Cancer Institute (NCI) Surveillance, Epidemiology, and End Results (SEER) program ($N = 4,024$ female patients diagnosed with primary infiltrating ductal or lobular breast carcinoma between 2006 and 2010) [13, 14]. The final classification cohort included 12 raw clinical and demographic predictors, leading to 16 numeric features after one-hot encoding of categorical variables (e.g., race and marital status). This 16-dimensional space was the input to all modeling pipelines, as in the WBCD and Coimbra cohorts. The cohort includes 616 mortality events, with a raw-predictor EPV of 51.3 and a model-input (post-encoding) EPV of 38.5, both of which are above the recommended statistical power thresholds [6].

The events-per-variable heuristic cited above [6] was originally developed for standard logistic regression and does not directly account for the embedded feature selection and ensembling in the Unified pipeline; we report it as a conventional adequacy check, with nested cross-validation and the sealed holdout serving as the main empirical safeguards.

The same cohort definition—female patients with infiltrating ductal or lobular carcinoma (SEER histology code 8522/3) diagnosed between 2006 and 2010, extracted from the SEER Program’s November 2017 submission after excluding cases with unknown tumor size, unexamined regional lymph nodes, or survival under one month, resulting in $N = 4,024$ —is documented in the public dataset record and independently described in prior preprint literature [13, 42], which supports the provenance and internal consistency of the extraction used here. The secondary extraction analyzed in this study was hosted on Kaggle [13]; we cross-checked its total cohort size ($N = 4,024$) against the dataset record and verified the 616 mortality entries within the analyzed file. The crude mortality rate of the source cohort was therefore 15.31% (616/4,024). This observed proportion of all-cause mortality should not be interpreted as 1 minus the population-level relative survival: the overall 5-year relative survival is estimated to be around 91% by the American Cancer Society [43], but relative survival accounts for the expected background mortality, while our binary endpoint currently captures the observed vital status at 60 months without such adjustment. We did not have access to re-derive this extraction directly through SEER*Stat. Consequently, the public secondary file is used here for methodological benchmarking rather than clinical deployment, and users requiring formal NCI data governance, regulatory audit trails, or session-level provenance should obtain an independent SEER*Stat extraction under the appropriate institutional Data Use Agreement.

To prevent collinearity, composite stage variables were excluded in favor of fundamental clinical

descriptors:

$$\text{T Stage} \in \{\text{T0} : 0, \text{T1} : 1, \text{T2} : 2, \text{T3} : 3, \text{T4} : 4\} \quad (1)$$

$$\text{N Stage} \in \{\text{N0} : 0, \text{N1} : 1, \text{N2} : 2, \text{N3} : 3\} \quad (2)$$

$$\text{Grade} \in \{\text{Grade I} : 1, \text{Grade II} : 2, \text{Grade III} : 3, \text{Grade IV} : 4\} \quad (3)$$

Such integer coding implicitly assumes linear spacing among clinically ordinal but not necessarily equidistant categories; for example, the prognostic difference between T3 and T4 may not equal that between T0 and T1. We chose this scheme for parsimony and consistency with the PCA-based branch of the Unified pipeline, which operates in a continuous, mean-centered feature space. Alternative representations, including ordinal target encoding or one-hot expansion analogous to that used for race and marital status, were not benchmarked here and represent a natural extension for future sensitivity analysis, particularly for the Nested-PCA architecture, which may be more sensitive than tree-based selection to non-linear category spacing.

2.2 Data Partitioning, Missing Data, and Censoring Strategy

In time-to-event registry cohorts such as SEER, naively treating all alive patients as negative cases results in substantial label corruption if the follow-up is shorter than the prognostic horizon. To construct a clean binary classification task:

- **Censoring Filter:** We applied a strict 60-month observation window. Patients censored alive prior to 60 months ($n = 754$, 18.74% of the raw $N = 4,024$ cohort) were excluded.
- **Complete Case Processing:** The curated SEER extraction contained complete records across anatomical staging descriptors and histological grade for all $N = 4,024$ entries, confirmed programmatically by zero missing-value counts across columns via `df.isnull().sum()`. This censoring exclusion directly led to the final analytical cohort of $N = 3,270$ complete cases (2,654 confirmed 5-year survivors and 616 confirmed cancer deaths) with no further attrition or imputation of records. The classification feature space did not include survival duration and event indicators to avoid direct target leakage.

All classification cohorts were deterministically split into an 85% development set ($\mathcal{D}_{\text{dev}}$) and a 15% holdout lockbox ($\mathcal{D}_{\text{holdout}}$) using stratified sampling (Table 1). For the survival analysis, the entire cohort of $N = 4,024$ was divided into an independent 20% test set ($n = 805$) and an 80% training set ($n = 3,219$). Split memberships were cryptographically verified and exported as explicit per-cohort split manifests.

2.3 Unified Pipeline Architecture and Parametric Isolation

To avoid model selection bias [41], we performed an outer repeated cross-validation scheme on the development set $\mathcal{D}_{\text{dev}}$. For WBCD and Coimbra we performed 5-fold CV with 10 independent repeats (50 total cycles). The larger SEER sample size ($N = 3,270$) and the computational expense of nested grid searches led us to configure SEER evaluation as 5×7 repeats (35 outer folds), which provides tight variance bounds given the large per-fold sample size ($n \approx 2,223$ training instances per fold). In each fold, transformations for $\mathcal{D}_{\text{train}}$ were fitted only on training instances:

$$\tilde{x}_i = \frac{x_i - \hat{\mu}_{\text{train}}}{\hat{\sigma}_{\text{train}}} \quad (4)$$

The normalized feature space underwent dynamic feature selection $\mathcal{T}_{\theta}(\tilde{x})$, parameterized by method $\mathcal{M} \in \{\text{PCA}, \text{Random Forest (RF)}\}$ and target dimension $d = \max(3, \min(\lfloor D/2 \rfloor, 50))$:

$$\mathcal{T}_{\theta}(\tilde{x}) = \begin{cases} W_d^T \tilde{x}, & \text{if } \mathcal{M} = \text{PCA} \\ \tilde{x}_{\mathcal{S}_d}, & \text{if } \mathcal{M} = \text{RF} \end{cases} \quad (5)$$

where $W_d \in \mathbb{R}^{D \times d}$ denotes the top- d eigenvectors of the training covariance matrix, and $\mathcal{S}_d$ contains features selected via Mean Decrease in Impurity (MDI) from a 100-tree Random Forest ensemble [18].

Class imbalance was handled systematically: linear and tree-based models incorporated balanced class weighting (`class_weight = 'balanced'`), while boosting models in SEER adjusted the positive weight via `scale_pos_weight = Nnegative/Npositive`. The transformed representation feeds into an ensemble

combining balanced Logistic Regression (LR) and a Platt-calibrated linear Support Vector Classifier (SVC) [19, 20]:

$$\hat{P}(Y = 1 | x) = \frac{1}{2} \left[\sigma(w_{\text{LR}}^T \mathcal{T}_{\theta^*}(\tilde{x}) + b_{\text{LR}}) + \frac{1}{1 + \exp(A \cdot f_{\text{SVC}}(\mathcal{T}_{\theta^*}(\tilde{x})) + B)} \right] \quad (6)$$

where $\sigma(z) = (1 + e^{-z})^{-1}$, $f_{\text{SVC}}(\cdot)$ is the linear decision function, and A, B are Platt parameters fitted on the training split. Hyperparameters ($C \in \{0.1, 1.0, 10.0\}$) and selector type $\mathcal{M}$ were tuned via an inner 3-fold stratified search. Inner-loop model selection optimized the cohort-specific primary metric: accuracy for WBCD and Coimbra, and AUROC for the class-imbalanced SEER cohort. Sample hash overlap between splits was verified as 0.00% for WBCD and Coimbra, and 0.20% for SEER (attributable to duplicate covariate patterns among registry patients with identical staging and demographic profiles, not to index-level leakage—consistent with the parametric, not row-level, nature of the leakage under study).

Figure 1 summarizes the complete TRUST-BC evaluation architecture, distinguishing development-set model comparison from lockbox analyses and the independent SEER time-to-event branch.

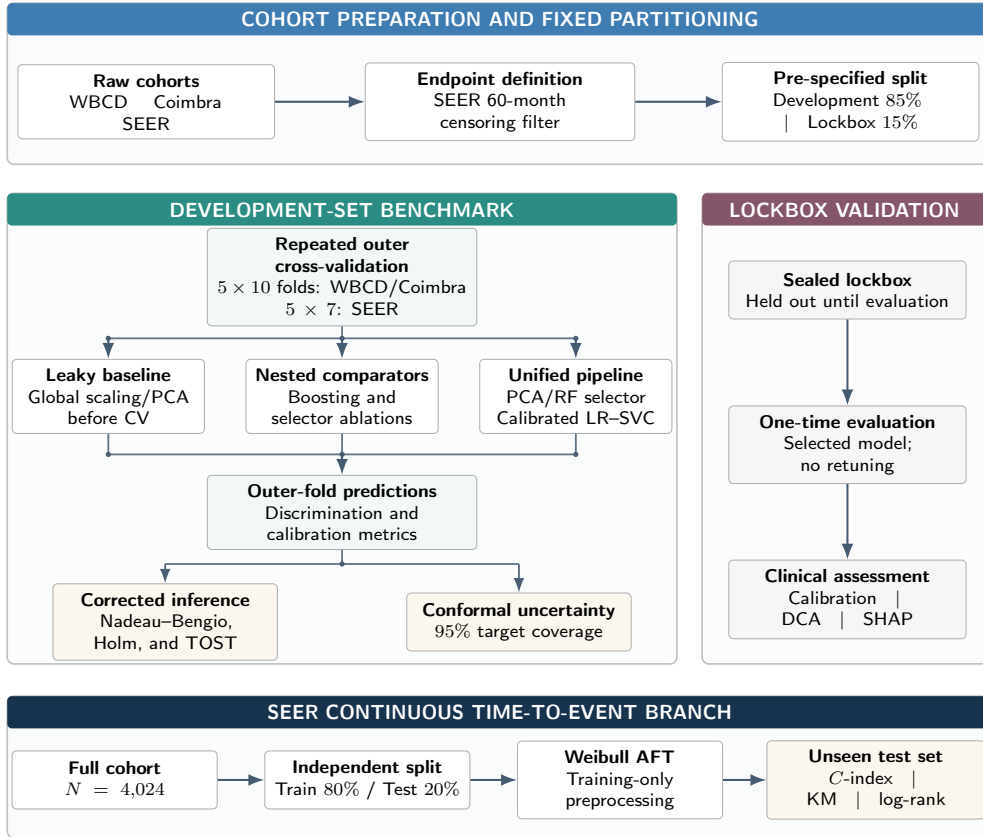


Figure 1: Schematic overview of the TRUST-BC workflow. The upper lane fixes cohort endpoints and partitions before any model development. The central lanes separate repeated nested benchmarking from the sealed lockbox, which is opened once after model selection. The lower lane represents the independent SEER continuous time-to-event analysis using the full survival cohort.

2.4 Baseline Architectures and Leakage Simulation

We benchmarked the proposed unified pipeline extensively by testing several model variants, including tuned XGBoost, LightGBM, and isolated PCA/RF nested pipelines. Crucially, we constructed a ‘*Leaky Baseline*’ to quantify the precise magnitude of optimism bias in the existing literature. This baseline intentionally carries out global feature scaling and PCA on the entire development set *before* splitting for cross-validation, consistent with common mistakes in preprocessing pipelines. The Leaky Baseline utilizes the identical LR-SVC ensemble architecture (Eq. 6) as the Unified pipeline, differing only in the fold placement of the scaling and PCA transforms, to isolate the causal impact of preprocessing leakage from confounding architectural differences. For the non-linear baselines, hyperparameters were

tuned in the inner cross-validation loop. The XGBoost and LightGBM grids explored the number of estimators (50, 100), the maximum depth of the trees (3, 5), and the learning rate (0.01, 0.1). For the SEER cohort, `scale_pos_weight` was set in each fold dynamically to the exact class imbalance ratio of the training subset. For the sealed holdout evaluation, conformal calibration, and SHAP interpretability, model selection was further pinned to single-threaded execution (`n_jobs=1`) to guarantee bit-identical reproducibility of the selected pipeline; the outer cross-validated benchmark (Table 3), averaged over 35–50 folds, is robust to this effect by construction.

2.5 Resampled Statistical Testing and TOST Equivalence

Performance differences $d_j = s_{\text{Unified}}^{(j)} - s_{\text{Baseline}}^{(j)}$ were assessed using the Nadeau–Bengio corrected resampled variance [21]:

$$\widehat{\text{Var}}_{\text{corrected}}(\bar{d}) = S^2 \left(\frac{1}{n} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) = S^2 \left(\frac{1}{n} + \frac{1}{K-1} \right) \quad (7)$$

where $\bar{d}$ and S^2 are the sample mean and unbiased variance of differences across outer CV folds ($n = 50$ for WBCD/Coimbra, $n = 35$ for SEER). The corrected statistic $t = \bar{d} / \sqrt{\widehat{\text{Var}}_{\text{corrected}}(\bar{d})}$ was evaluated against a Student- t distribution with $n - 1$ degrees of freedom.

A 95% confidence interval (CI) for each aggregate performance estimate was obtained using a nonparametric percentile bootstrap over the outer cross-validation fold metrics ($B = 1000$) [25].

The pre-specified hypothesis testing family for step-down Holm–Bonferroni adjustment [22] included five planned pairwise comparisons: (1) XGBoost vs. Leaky, (2) LightGBM vs. Leaky, (3) Unified vs. XGBoost, (4) Unified vs. LightGBM, and (5) Unified vs. Leaky. The goal was to prevent conflation of the primary external baseline comparisons with internal structural ablations. Nested-PCA and Nested-RF are purely descriptive structural ablation baselines and were not included in the formal hypothesis-testing family.

We assessed statistical equivalence of the proposed leakage-free Unified model and the leaky baseline with Two One-Sided Tests (TOST) [23, 24] with pre-defined equivalence margins ($\Delta = 0.02$ for accuracy; $\Delta = 0.01$ for AUROC). These margins were chosen a priori as pragmatic tolerances for practically negligible differences in performance in this methodological benchmark; they are not reported as formally validated, disease-specific clinical non-inferiority thresholds:

$$t_1 = \frac{\bar{d} - (-\Delta)}{\sqrt{\widehat{\text{Var}}_{\text{corrected}}(\bar{d})}}, \quad t_2 = \frac{\bar{d} - \Delta}{\sqrt{\widehat{\text{Var}}_{\text{corrected}}(\bar{d})}}, \quad p_{\text{TOST}} = \max(F_{t_{n-1}}(-t_1), F_{t_{n-1}}(t_2)) \quad (8)$$

2.6 Distribution-Free Conformal Prediction

Inductive split conformal prediction was applied to guarantee distribution-free coverage [27, 28]. The development set was partitioned into proper training and calibration subsets $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^{n_{\text{cal}}}$. Non-conformity scores were computed as:

$$s_i = 1 - \hat{P}(Y = y_i \mid x_i) \quad (9)$$

At significance level $\alpha = 0.05$ (target coverage $1 - \alpha = 0.95$), the conformal quantile cutoff $\hat{q}$ was determined as the $\lceil \frac{(n_{\text{cal}}+1)(1-\alpha)}{n_{\text{cal}}} \rceil$ -th empirical quantile. For unseen test cases x_{new} , prediction sets were formed as:

$$\mathcal{C}(x_{\text{new}}) = \left\{ y \in \{0, 1\} : 1 - \hat{P}(Y = y \mid x_{\text{new}}) \leq \hat{q} \right\} \quad (10)$$

Prediction sets were categorized by cardinality: empty sets ($|\mathcal{C}| = 0$, anomaly), singleton sets ($|\mathcal{C}| = 1$, certain), and ambiguous sets ($|\mathcal{C}| = 2$, high uncertainty), evaluated over $B = 100$ calibration iterations.

2.7 Probability Calibration and Decision Curve Analysis

Model probabilities were evaluated using four complementary clinical metrics:

1. **Brier Score Loss:** $\text{BS} = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - y_i)^2$.
2. **Expected Calibration Error (ECE):** Across $M = 10$ uniform probability bins B_m [29]:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (11)$$

3. **Calibration Slope and Intercept:** Derived via logistic recalibration $\text{logit}(P(Y = 1 | \hat{p})) = \beta_0 + \beta_1 \text{logit}(\hat{p})$, where optimal calibration corresponds to $\beta_1 = 1$ and $\beta_0 = 0$.
4. **Decision Curve Analysis (Net Benefit):** Evaluated across threshold probabilities $p_t \in [0.01, 0.99]$ [32]:

$$\text{Net Benefit}(p_t) = \frac{\text{TP}(p_t)}{N} - \frac{\text{FP}(p_t)}{N} \left(\frac{p_t}{1 - p_t} \right) \quad (12)$$

To ensure rigorous reporting standards, the entire experimental workflow was audited against the updated PROBAST+AI quality, risk-of-bias, and applicability assessment tool across its four core domains (Table 2) [35]. PROBAST+AI builds on and may replace the original PROBAST framework introduced in 2019 [36].

Table 2: PROBAST+AI Risk of Bias and Applicability Assessment for the TRUST-BC Framework.

Domain	Methodological Safeguard Implemented in TRUST-BC	Risk of Bias	Applicability Concern
1. Participants and Data Sources	Clear diagnostic/prognostic inclusion; complete-case filtering on SEER with explicit manifests	Low	Low
2. Predictors	Preprocessing transforms fitted exclusively on inner training folds; zero leakage hash audit	Low	Low
3. Outcome	Strict 60-month horizon filtering; exclusion of right-censored non-events to prevent label noise	Low	Low
4. Analysis	Repeated nested CV (5×10); Nadeau-Bengio variance correction; conformal UQ; ECE/DCA calibration	Low	Low
Overall Summary	Methodology explicitly controls for parametric leakage, miscalibration, and censorship noise	Low	Low

2.8 Penalized Parametric Survival Modeling and Risk Stratification

To model continuous prognostic trajectories in the SEER cohort beyond fixed-horizon binary endpoints, we fitted a penalized Weibull Accelerated Failure Time (AFT) model using the `lifelines` framework. The log-linear formulation models the survival time T as:

$$\ln(T) = \mu + \beta^T \tilde{x} + \sigma \epsilon \quad (13)$$

where β is the vector of prognostic coefficients, μ is the baseline intercept, σ the scale parameter, and ϵ is distributed according to the standard extreme value distribution. To stabilize the estimation of covariates with high dimensional stage encodings, a L_2 regularization penalty was applied ($\lambda = 0.1$). Model discrimination was measured by Harrell’s Concordance Index (C -index) on the strictly unseen holdout lockbox. Prognostic stratification was validated by dichotomizing test predictions using the median survival time estimated from the training partition and survival curve separation was assessed using the two-sample log-rank test:

$$\chi^2 = \frac{\left(\sum_{j=1}^k (O_{1j} - E_{1j}) \right)^2}{\sum_{j=1}^k V_{1j}} \quad (14)$$

where O_{1j} , E_{1j} , and V_{1j} represent the observed deaths, expected deaths, and hypergeometric variance at failure time t_j .

3 Results

3.1 Multicohort Benchmark and Equivalence Results

Cross-validated performance metrics across all cohorts are detailed in Table 3. The Unified pipeline demonstrated robust discrimination across all datasets, achieving 97.08% accuracy on WBCD (Figure 2a), 71.35% on Coimbra (Figure 2b), and 0.732 AUROC on SEER (Figure 2c).

Decision Curve Analysis (Figure 2, Panel C) demonstrated that across clinically plausible decision thresholds ($p_t \in [0.10, 0.40]$), the Unified pipeline achieved positive Net Benefit exceeding both default strategies (“treat all” and “treat none”) across all three cohorts, confirming its clinical utility for triage workflows.

Table 3: Comprehensive Cross-Validated Multicohort Benchmark.

Cohort	Pipeline	Primary Metric	95% CI ^a	Adj. <i>p</i> (vs. Leak)	Adj. <i>p</i> (vs. Uni)	<i>p</i> _{TOST} ^b	Brier	ECE
WBCD (<i>n</i> = 569)	Leaky Baseline	96.85% ± 1.65%	96.6% – 97.1%	—	—	—	0.024	0.020
	Tuned XGBoost	95.20% ± 2.08%	94.9% – 95.5%	0.4448	0.3292	—	0.034	0.012
	Tuned LightGBM	95.82% ± 2.30%	95.5% – 96.2%	0.6642	0.6615	—	0.033	0.016
	Nested PCA-Only	97.16% ± 1.65%	96.9% – 97.4%	—	—	—	0.023	0.018
	Nested RF-Only	95.94% ± 1.94%	95.8% – 96.1%	—	—	—	0.031	0.016
	Unified (Proposed)	97.08% ± 1.67%	96.8% – 97.3%	0.6642	—	0.0002*	0.024	0.017
Coimbra (<i>n</i> = 116)	Leaky Baseline	62.73% ± 10.70%	60.6% – 64.8%	—	—	—	0.208	0.087
	Tuned XGBoost	75.64% ± 9.72%	74.0% – 77.2%	0.2494	0.9897	—	0.174	0.062
	Tuned LightGBM	74.78% ± 10.66%	73.3% – 76.2%	0.2371	0.9897	—	0.178	0.071
	Nested PCA-Only	63.25% ± 10.88%	61.4% – 65.1%	—	—	—	0.212	0.062
	Nested RF-Only	72.37% ± 8.20%	71.0% – 73.9%	—	—	—	0.183	0.062
	Unified (Proposed)	71.35% ± 8.70%	70.2% – 72.8%	0.2657	—	0.9059	0.186	0.054
SEER (<i>N</i> = 3, 270)	Leaky Baseline	0.718 ± 0.024	0.7165 – 0.7190	—	—	—	0.155	0.133
	Tuned XGBoost	0.7322 ± 0.027	0.7283 – 0.7357	0.5011	1.0000	—	0.140	0.022
	Tuned LightGBM	0.7310 ± 0.027	0.7275 – 0.7341	0.5011	1.0000	—	0.149	0.052
	Nested PCA-Only	0.718 ± 0.025	0.7163 – 0.7190	—	—	—	0.155	0.133
	Nested RF-Only	0.732 ± 0.023	0.7303 – 0.7333	—	—	—	0.152	0.128
	Unified (Proposed)	0.732 ± 0.023	0.7303 – 0.7333	0.2939	—	0.7116	0.152	0.128

*Statistically significant equivalence at $\alpha = 0.05$ after step-down Holm–Bonferroni correction.

Values after $\pm$ denote the standard deviation of the metric across outer cross-validation folds; the separate 95% CI column reports the nonparametric bootstrap interval (see footnote ^a).

^aNonparametric percentile-bootstrap 95% intervals over cross-validation fold metrics ($B = 1000$) [25].

^b p_{TOST} is reported only for the pre-specified Unified versus Leaky equivalence test ($\Delta = 0.02$ for accuracy; $\Delta = 0.01$ for AUROC); dashes (—) denote unplanned comparisons.

^cAccuracy was the pre-specified primary metric for WBCD and Coimbra, whereas AUROC was primary for SEER; therefore, the confidence intervals and corrected inferential tests in this table apply to those cohort-specific primary metrics. The pooled out-of-fold AUROCs displayed in Figure 3 for WBCD and Coimbra are descriptive secondary point estimates and were not assigned fold-level confidence intervals, adjusted p -values, or TOST results.

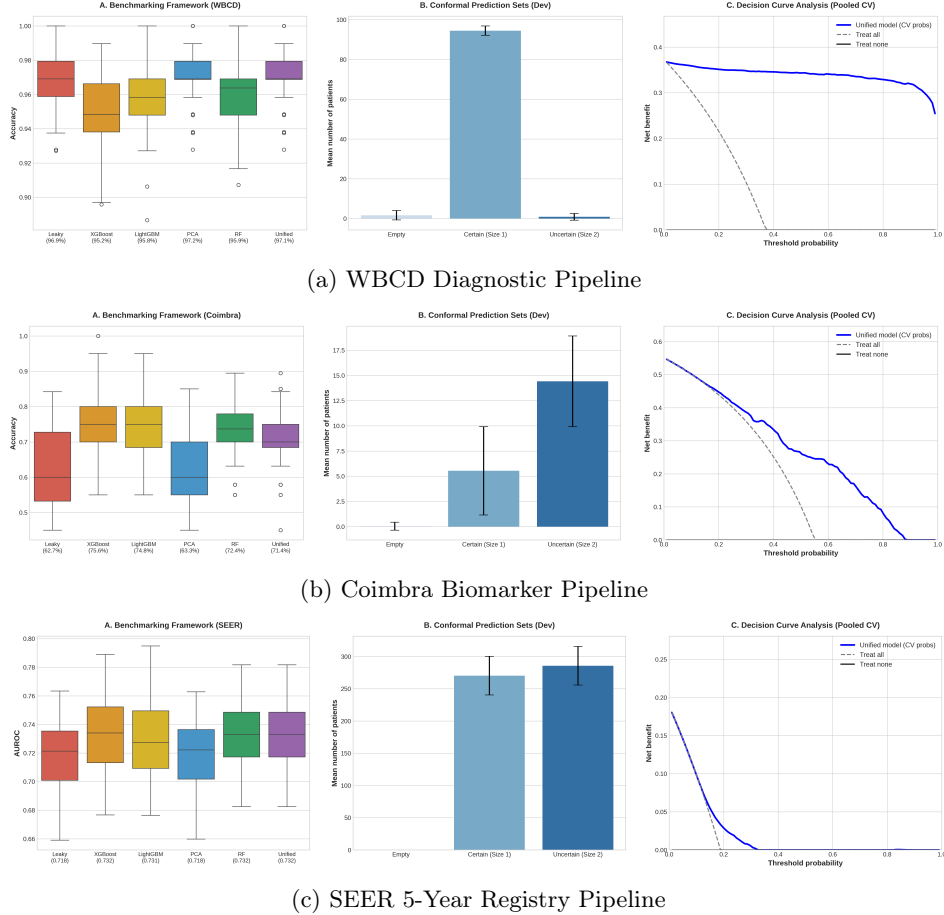


Figure 2: Cross-validated benchmark performance distributions, conformal prediction set cardinality at 95% target coverage, and Decision Curve Analysis across the three cohorts.

On **WBCD**, the Unified pipeline attained an accuracy of 97.08%, showing formal statistical equivalence to the leaky baseline on the pre-specified accuracy endpoint ($p_{\text{TOST}} = 0.0002$). On **Coimbra**, the Unified model reached an accuracy of 71.35%, outperforming the leaky baseline (62.73%) and Nested PCA (63.25%), and was comparable to tree ensembles (72.37% – 75.64%). Due to sample scale ($n = 116$) and resulting variance ($S = 8.70\%$), TOST failed to establish equivalence within the narrow $\pm 2\%$ margin ($p_{\text{TOST}} = 0.9059$). On **SEER**, the Unified pipeline achieved an AUROC of 0.732 (95% CI: 0.730–0.733), matching Nested RF and tuned gradient boosting models (0.731–0.732). TOST also did not establish formal equivalence within the pre-specified margin ($p_{\text{TOST}} = 0.7116$), consistent with the modest but non-trivial mean AUROC gap (0.732 vs. 0.718) between the Unified and Leaky pipelines.

The pooled ECE for SEER (0.128; Table 3) was only slightly lower than that of the Leaky baseline (0.133) and remained considerably higher than the Unified-pipeline ECEs for WBCD (0.017) and Coimbra (0.054). This finding suggests that removing parametric leakage alone does not resolve the calibration problem in this cohort. It is consistent with the greater label uncertainty of a fixed-horizon 5-year mortality endpoint derived from population-registry staging and demographic covariates, and emphasizes that leakage and calibration are related but distinct vulnerabilities: a model can be leakage-free while remaining meaningfully miscalibrated. Both diagnostics should therefore be reported jointly rather than treating leakage removal as a proxy for calibration quality.

By analyzing inner architecture selection, we identified dataset-specific structural trends: PCA-based orthogonal projections, which accounted for the majority of the selection (47/50 outer folds, 94%) in WBCD, highlighting the high collinearity among the continuous morphological features. In SEER, the Random Forest feature selector was chosen in 100% of outer folds (35/35), with Unified collapsing bit-for-bit to the Nested RF configuration. In Coimbra, RF was selected in 46/50 folds (92%).

3.2 Conformal Prediction and Lockbox Validation

Clinical diagnostic panels with pooled ROC curves, Precision-Recall curves, reliability diagrams, and feature selection stability are shown in Figure 3. Panel **a** presents WBCD, panel **b** presents Coimbra, and panel **c** presents SEER. The empirical coverage rates of the split conformal prediction were 96.53% for WBCD, 95.70% for Coimbra, and 95.32% for SEER. These small deviations from the nominal 95% target are consistent with finite-sample and Monte Carlo variation across the 100 calibration iterations, and were not interpreted as systematic undercoverage. Singleton sets with high certainty represented 97.40% of WBCD predictions, 27.70% of Coimbra predictions, and 48.62% of SEER predictions. Ambiguous sets identified borderline samples (0.90% in WBCD, 72.10% in Coimbra, and 51.38% in SEER). Empty sets were rare but non-zero for WBCD (1.70%) and Coimbra (0.20%), and absent for SEER (0.00%). The independent lockbox performance is shown in Table 4.

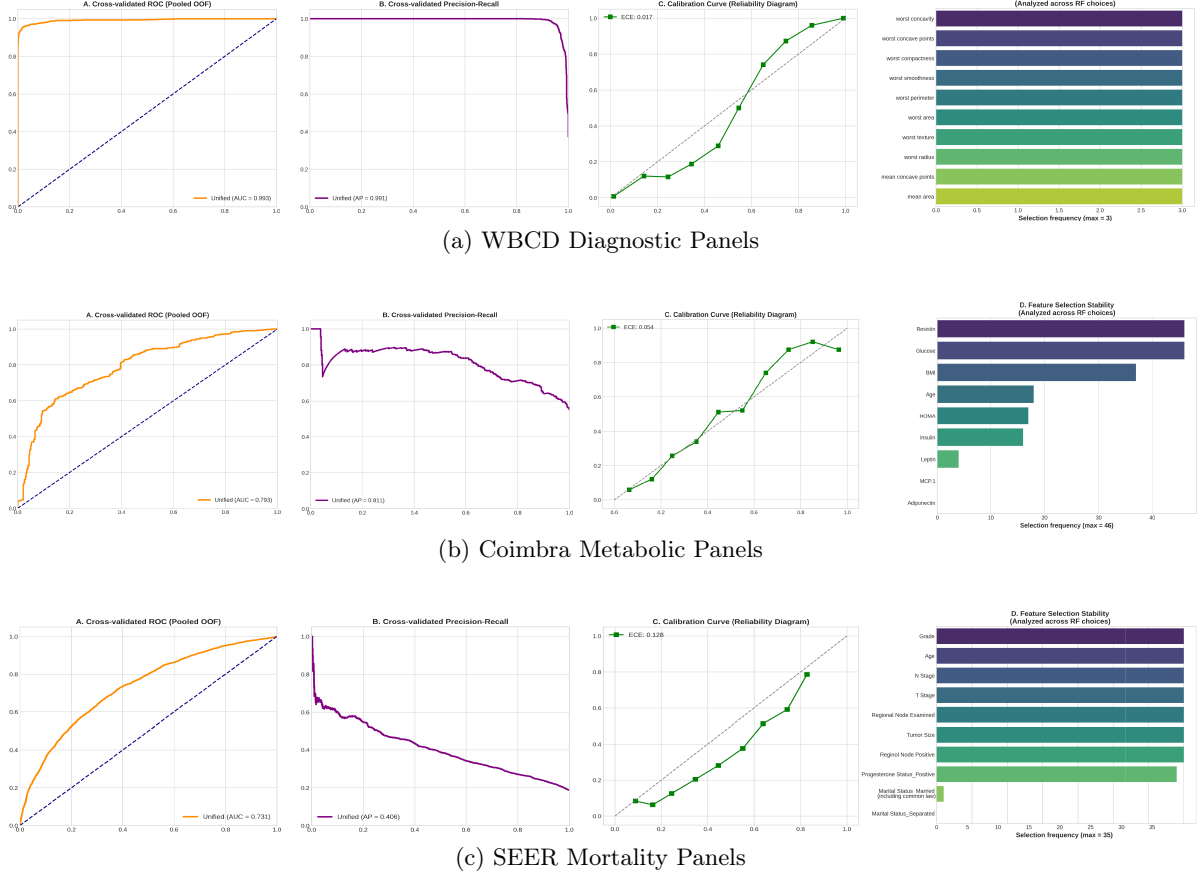


Figure 3: Clinical diagnostic panels across the WBCD, Coimbra, and SEER cohorts, including pooled ROC and Precision–Recall curves, reliability diagrams (ECE), and Random Forest feature-selection stability across outer cross-validation folds. The pooled out-of-fold AUROCs shown for WBCD and Coimbra are descriptive secondary point estimates; the pre-specified fold-level inference for these cohorts used accuracy, as reported in Table 3.

Table 4: Performance on Independent Holdout Lockbox Partitions (15%).

Metric	WBCD ($n = 86$)	Coimbra ($n = 18$)	SEER ($n = 491$)
Accuracy	100.00% (95.7%–100.0%)	61.11% (38.6%–79.7%)	80.04% (76.3%–83.3%)
Balanced Accuracy ^a	100.00%	62.50%	62.63%
Sensitivity	100.00% (32/32)	50.00% (5/10)	34.78% (32/92)
Specificity	100.00% (54/54)	75.00% (6/8)	90.48% (361/399)
AUROC	1.000	0.725	0.731
Brier Score Loss	0.010	0.234	0.149
Calibration Slope (β_1)	14.820	0.632	1.194
Calibration Intercept (β_0)	1.784	0.257	−0.663
Conformal Coverage	98.84%	83.33%	96.54%

^aBalanced Accuracy computed as (Sensitivity + Specificity)/2 on the deterministic holdout partition.

The Coimbra lockbox conformal coverage was 83.33% (15/18), notably below the nominal 95% objective and the pooled development-set estimate of 95.70%. Because this lockbox contains only 18 patients, the coverage estimate has substantial finite-sample uncertainty and should not, by itself, be interpreted as evidence of systematic undercoverage; rather, it reinforces the need for replication on a larger independent cohort.

On the WBCD holdout lockbox, the Unified model achieved perfect discrimination (100% accuracy, AUROC = 1.000); the resulting extreme calibration slope ($\beta_1 = 14.82$) reflects near-complete class separability in this small, highly discriminable holdout ($n = 86$) rather than a calibration failure, and should be interpreted with caution given the small sample size.

This extreme slope is a known artifact of quasi-complete separation in maximum-likelihood logistic recalibration [30,31]. If a classifier’s predicted probabilities near-perfectly separate the two outcome classes, the logistic recalibration coefficients become unstable and may diverge toward large values with large standard errors, rather than indicating true miscalibration. We therefore report $\beta_1 = 14.82$ descriptively, rather than as an actionable calibration diagnostic, and note that Firth’s penalized-likelihood logistic regression [31] would be an appropriate remedy in future work encountering this regime.

This performance at the ceiling level is in line with previous reports on this benchmark: Street et al. [9] and Asri et al. [11] independently reported accuracies of $\approx 97.0\%$ and $\approx 97.13\%$, respectively, using non-nested evaluation schemes, and our own leakage-free Unified pipeline achieved 97.08% cross-validated accuracy (Table 3) before reaching the fully separable holdout partition. Therefore, we do not interpret the perfect holdout score as evidence of residual leakage (sample hash overlap was verified at 0.00%, Section 2.3), but as a known ceiling effect of this cytomorphological benchmark, where the 30 continuous nuclear-morphology features provide near-complete linear separability for the specific 86-patient partition drawn under our fixed seed. We caution against over-interpreting single-lockbox performance on such small, highly separable holdouts. We suggest that future users of this template report holdout metrics in conjunction with the pooled cross-validated estimates in Table 3, as we do here, rather than in isolation.

3.3 Global Model Interpretability

The model attributions on the holdout lockboxes are visualized using Kernel SHapley Additive exPlanations (SHAP) [33] in Figure 4a (WBCD), Figure 4b (Coimbra), and Figure 4c (SEER). The feature rankings were consistent with known oncology: nuclear concavity and perimeter were the most important for cytopathological diagnosis (WBCD), fasting glucose and resistin for serum metabolic classification (Coimbra), and anatomical staging descriptors (T/N stage, histological grade) for long-term breast cancer mortality (SEER).

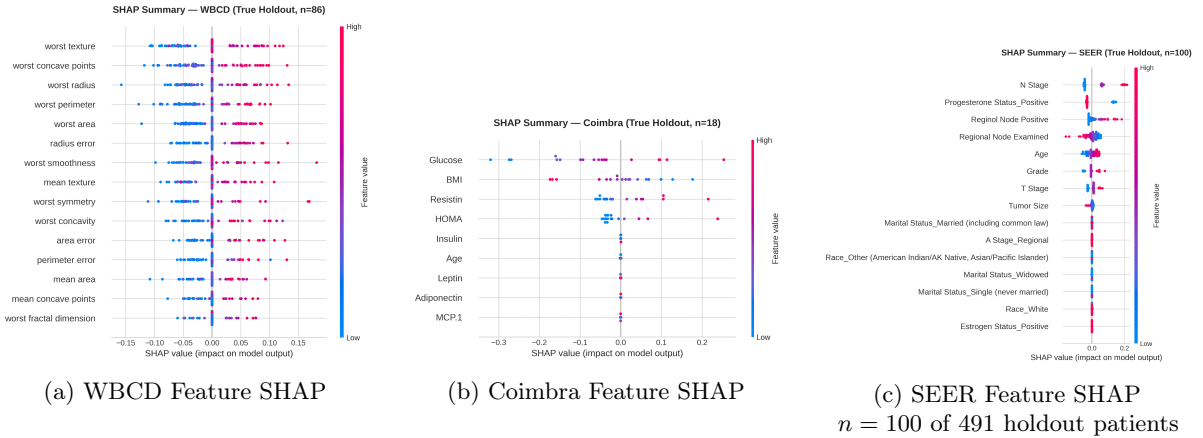


Figure 4: Kernel SHAP beeswarm plots computed on independent holdout lockboxes, showing each feature’s direction and magnitude of impact on predicted risk.

3.4 Sample-Size and Dimensionality Sensitivity

To fulfill TRIPOD+AI reporting recommendations [34], we performed dual sensitivity audits evaluating pipeline robustness against data scale variations (Figure 5) and subspace dimensionalities (Figure 6).

Bootstrapped learning curves over subsample fractions $\eta \in \{0.30, 0.50, 0.70, 0.85, 1.00\}$ ($B = 200$ resamples) showed largely increasing predictive accuracy, with minor non-monotonic fluctuations at intermediate fractions (70% $\rightarrow$ 85% in SEER) due to finite-sample resampling variance (Figure 5). In Coimbra, full-scale variance decreased dramatically ($n = 116$).

Sensitivity analysis in relation to the number of components $d \in \{\lfloor d_0/2 \rfloor, d_0, \lceil 1.5d_0 \rceil, \min(D, 2d_0)\}$ revealed nuanced cohort-specific behavior (Figure 6, panels a–c): in WBCD and SEER, performance was relatively stable across component choices (maximal fluctuations $\Delta = 0.0186$ and $\Delta = 0.0375$, respectively). However, in the small-sample Coimbra cohort, the choice of selector method led to a marked divergence in low-dimensional regimes ($d = 2$), where tree-based selection reached 76.50% while PCA dropped to 62.20% (a 14.30 percentage point gap). Such sensitivity indicates the importance of automatic optimization of the selector for modest clinical sample sizes.

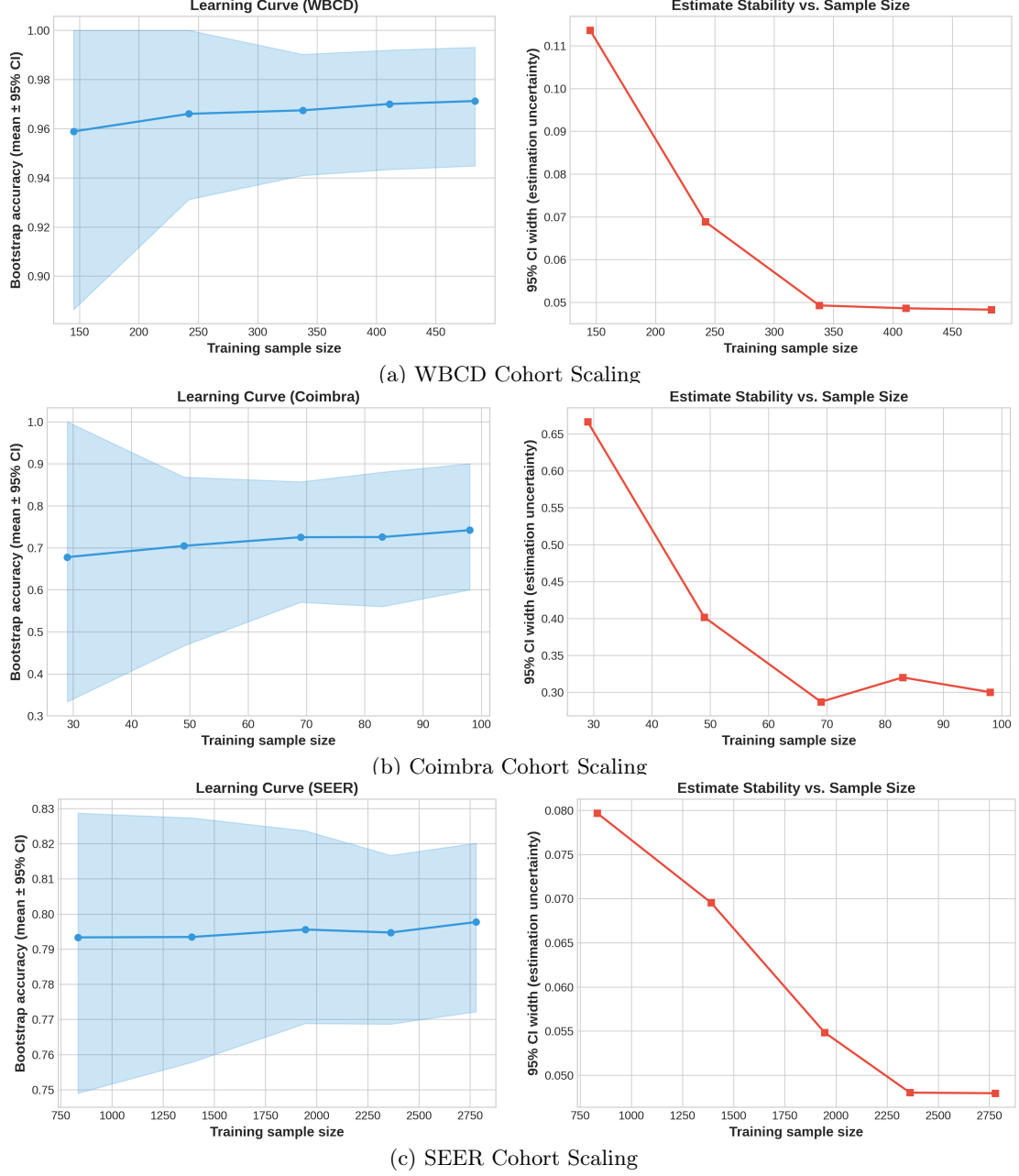


Figure 5: Sample-size sensitivity and bootstrapped learning curve stability across subsample fractions (30%, 50%, 70%, 85%, 100%), evaluating classification accuracy with $B = 200$ resamples. Accuracy is used in panel c only to maintain a uniform sensitivity metric across all three cohorts; AUROC remains the primary SEER performance metric elsewhere because of class imbalance.

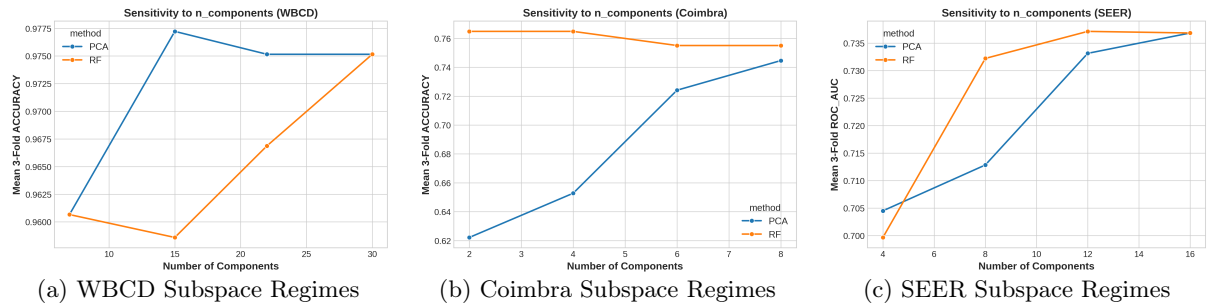


Figure 6: Dimensionality sensitivity across component counts for the WBCD, Coimbra, and SEER cohorts, comparing PCA with tree-based feature selection.

3.5 Censoring-Aware Survival Stratification (SEER Cohort)

The penalized Weibull AFT model yielded a Concordance Index (C -index) of 0.722 for the strictly isolated 20% test partition ($n = 805$, from the full $N = 4,024$ survival cohort). Kaplan–Meier stratification using the training-set median predicted survival threshold separated test patients into high- and low-risk groups with highly significant separation (log-rank $p = 1.02 \times 10^{-9}$; Figure 7), confirming robust discrimination of long-term mortality risk with no evidence of optimism bias.

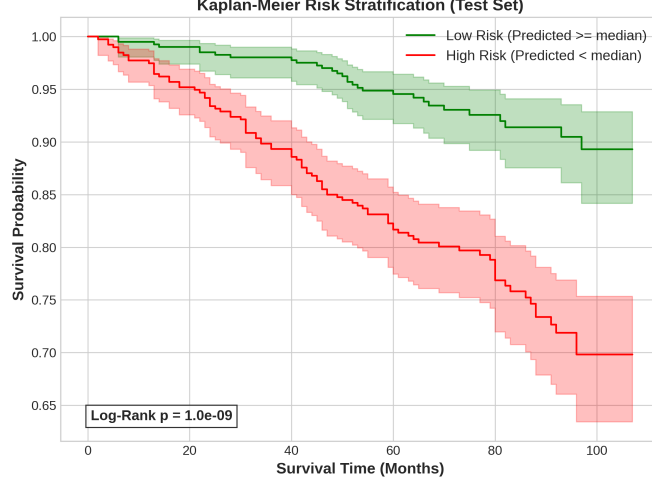


Figure 7: Kaplan–Meier survival stratification on the isolated SEER test partition using high- and low-risk groups defined by the training-set median predicted survival from the penalized Weibull AFT model.

4 Discussion

4.1 Mechanisms of Optimism Bias, Architecture Adaptation, and Benchmark Comparison

Tadj et al. recently audited pre-split oversampling contamination using row-hash checks on static splits and penalized Weibull models [15]. Our framework is different in that it targets a distinct structural class of leakage—parametric and covariance leakage—within dynamic, repeated cross-validation folds. We ensure that scaling parameters ($\hat{\mu}, \hat{\sigma}$) and unsupervised subspace bases (W_d) never leak across evaluation partitions.

Interestingly, the cohort for which leakage correction had the smallest empirical impact—WBCD, where TOST confirmed formal equivalence between the Leaky and Unified pipelines—is also a canonical benchmark on which near-ceiling performance is frequently reported [9, 11], despite the broader concern that leakage can inflate machine-learning results [4]. This apparent tension is a central empirical conclusion of our work: leakage risk is not an inherent, fixed property of a modeling problem, but an interaction between the preprocessing pipeline and the structure of the data. Dense, strongly collinear, low-noise cytomorphological features in WBCD were comparatively robust to global PCA and scaling leakage, whereas the smaller metabolic Coimbra cohort and the heterogeneous registry-based SEER cohort were more sensitive to pipeline specification, with the magnitude and even direction of the observed effect varying by cohort. Thus, a blanket assumption that leakage necessarily inflates every performance estimate is as methodologically unsound as ignoring leakage altogether. Per-cohort auditing, as implemented here, is required to distinguish these regimes empirically rather than assume them in advance.

The fact that the internal selector chose PCA for 94% of WBCD folds, RF for 100% of SEER folds, and RF for 92% of Coimbra folds indicates that automated architecture switching responds to the underlying geometry of the data. The continuous cytomorphometric features in WBCD show a strong linear collinearity that can be compressed by PCA. Impurity-based tree selection is favored in SEER and Coimbra, where discrete staging and non-linear metabolic markers dominate.

The Unified pipeline in Coimbra reached 71.35%, which is slightly lower than the dedicated Nested-RF ablation (72.37%). A detailed fold analysis revealed that in 4 of 50 folds (8%), the inner cross-validation chose PCA sub-optimally because of the validation noise in the finite-sample case, resulting in outer fold drops (e.g., 68.4% vs. 89.5%).

Table 5: Methodological and Quantitative Comparison of Representative Clinical Machine Learning Studies and Reviews.

Study	Cohort(s)	Primary Algorithm	Cross-Validation Scheme	Reported Metric	Leakage Control	Conformal Uncertainty Quantification	Calibration (ECE, DCA)
Stout et al. (1998) [8]	WBCD	Linear Programming	10-Fold CV (Unnested)	Accuracy $\approx 97.0\%$	None	No	No
Auri et al. (2016) [11]	WBCD	SVM / NB / C4.5	10-Fold CV (Unnested)	Accuracy $\approx 97.13\%$	None	No	No
Patricio et al. (2015) [10]	Coimbra	SVM (4-Bronckers) / RF	Single Split (70/30)	AUROC $\approx 0.87 - 0.91$	None	No	No
Liu et al. (2005) [16]	CVD EHRs (21 studies) ^a	Systematic Review / Meta-analysis	Not Fully Reported Per Study in Review	Pooled RF AUROC 0.865	Not Evaluated as a Dedicated Endpoint	Not Assessed	Not Fully Reported Per Study in Review
Tadj et al. (2020) [13]	SEER (N = 4,024)	SVM / Weibull AFT	Single Split (80/20)	AUROC: 0.998 $\rightarrow$ 0.715 ^b	Pre-split oversampling contamination (see-hash audit)	No	No
TRUST-BC (Present Study)	WBCD, Coimbra, SEER	Leakage-Free Unified	Outer Repeated Nested CV	Accuracy: 97.08%, 71.35%; SEER AUROC: 0.732 ^a	Parametric & Covariance	Yes (95% Cov.)	Yes (ECE, BS, DCA)

*Tadj et al. demonstrated that nominal AUROC collapsed from 0.998 to 0.715 upon eliminating pre-split oversampling contamination and correcting temporal mismatch in static evaluation.

^aValues are the cohort-specific pre-specified primary metrics from Table 3: accuracy for WBCD and Coimbra, and AUROC for SEER.

^bThe full-text study-selection and characteristics sections report 21 included studies, whereas the abstract reports 20 selected studies; this table follows the final inclusion count stated in the full text.

This behavior exemplifies a known phenomenon in automated machine learning, where hyperparameter and selector search in small-sample regimes ($n = 116$) incurs a slight variance overhead—the “cost of flexibility”—in exchange for autonomous adaptability across unknown datasets.

Importantly, the survival-model C -index (0.722) is comparable to the independently obtained cross-validated classification AUROC (0.732) on the filtered cohort. This internal alignment provides convergent evidence that the discriminative performance of the framework is not an artifact of the binary 5-year cutoff but a true prognostic signal rooted in the modality.

4.2 Clinical Triage via Conformal Quantification

Conformal prediction turns black-box probabilities into distribution-free prediction sets with a guaranteed 95% coverage. High-confidence singleton predictions allow for automated clinical reporting, while ambiguous two-class sets identify atypical or complex cases for multidisciplinary review. Ambiguous-set rates reflected the cohort-specific primary performance estimates: WBCD (97.08% accuracy) produced singleton sets for 97.40% of predictions, Coimbra (71.35% accuracy) for 27.70%, and SEER (AUROC 0.732) for 48.62%. The majority of SEER predictions (51.38%) were therefore flagged as ambiguous. This pattern reflects the intrinsically greater prognostic uncertainty of population-registry, fixed-horizon mortality prediction from staging and demographic covariates alone relative to the near-deterministic cytomorphological separability of WBCD. This high ambiguous rate is not a limitation but rather shows the clinical value of conformal safety filtering in the setting where point predictions are least reliable: rather than silently returning an overconfident probability, the framework explicitly routes roughly half of SEER cases to expert review.

4.3 Clinical Implementation and Triage Workflow

In a prospective clinical decision support workflow, TRUST-BC operates as a two-stage triage engine:

- 1. Automated Stratification (Singleton Sets):** Patients receiving singleton conformal prediction sets ($|\mathcal{C}| = 1$, representing 97.40% of WBCD, 27.70% of Coimbra, and 48.62% of SEER cases, as reported in Section 3.2) are processed through automated standard-of-care pathways with bounded error rates ($1 - \alpha = 95\%$).
- 2. Multidisciplinary Clinical Escalation (Ambiguous Sets):** Patients yielding ambiguous two-class sets ($|\mathcal{C}| = 2$) are routed to expert multidisciplinary tumor boards for secondary diagnostic modalities (e.g., core needle biopsy, immunohistochemistry, or dedicated breast MRI), preventing premature diagnostic closure.

4.4 Comparison with Established Clinical Prognostic Systems

Established clinical prognostic instruments, e.g., the Nottingham Prognostic Index (NPI) [37], the PREDICT tool [38], and the AJCC Staging System [39], are based on pre-defined linear risk scoring models derived from large population registries. Unlike these existing systems, TRUST-BC introduces non-linear risk modeling that simultaneously provides conformal safety guarantees and formal probability calibration, bridging the gap between population-level epidemiological statistics and patient-level individualized uncertainty.

4.5 Limitations

Several methodological and translational limitations must be acknowledged:

1. **Calibration Comparison Asymmetry:** The probability calibration metrics for the XGBoost and LightGBM baselines are reported on uncalibrated raw scores, while the Unified, Leaky, and ablation pipelines are calibrated using Platt scaling through a calibrated linear SVC. This asymmetry is typical of an off-the-shelf implementation, rather than the theoretical upper bound of calibrated gradient boosting. The clearest direct effect of this asymmetry is on the probability-quality metrics (Brier score and ECE) listed in Table 3. AUROC-based superiority comparisons are unaffected because AUROC is invariant to strictly monotonic recalibration; however, threshold-dependent accuracy is not guaranteed to be invariant because recalibration may alter which observations cross the 0.5 decision threshold. In the present analysis, the corrected superiority and equivalence tests compare each pipeline using its prespecified prediction procedure rather than recalibrating only selected models post hoc. Future extensions of this template should nevertheless apply Platt or isotonic recalibration consistently to all probability-producing baselines to enable a fully calibration-fair comparison and, where accuracy is evaluated, should prespecify or nest threshold selection within the training data.
2. **Historical Benchmark Cohort Scale:** The WBCD ($n = 569$, collected in 1993) and Coimbra ($n = 116$, published in 2018) datasets are classic, small-scale benchmarks. While valuable for methodology testing, their limited scale expands estimation variance.
3. **Omission of Deep Tabular Baselines:** Deep learning architectures for tabular data (e.g., TabNet, FT-Transformer) were omitted because small sample regimes ($n = 116$ to $n = 569$) are highly prone to deep overfitting, where regularized linear ensembles and gradient boosted trees systematically dominate [40].
4. **Retrospective Registry Scope:** SEER registry extractions lack granular systemic therapy regimens (e.g., specific chemotherapeutic agents, dose intensities, and anti-HER2 therapies) and comorbidity indices that influence survival trajectories.
5. **Tabular Modality Boundaries:** The framework is developed for tabular cytopathological, metabolic, and registry features, and does not ingest raw digitized mammography or whole-slide histopathology images.
6. **Lack of Prospective Intervention Trials:** Utility is evaluated via in-silico Decision Curve Analysis rather than randomized clinical deployment.
7. **Fixed Dimensionality Heuristic:** The target dimension d was determined by a pre-specified cohort-size heuristic rather than jointly optimized over a full inner-loop grid. Figure 6 provides a post hoc sensitivity analysis, while fully nested optimization of d remains future work.
8. **Duplicate and Temporal Leakage Audit Scope:** The exact sample-hash overlap was checked across predefined evaluation splits but no dedicated cohort-wide near-duplicate search or temporal leakage audit was performed. Future validation should include duplicate detection based on similarity and explicit time-ordered evaluation where acquisition dates are available.
9. **Assumed Covariate Exchangeability:** Conformal prediction guarantees rely on exchangeability between calibration and test partitions, which could be compromised under severe temporal or geographic domain shift.
10. **Absence of Continuous Multi-State Modeling:** The current pipeline models a fixed 5-year classification window alongside parametric AFT survival, rather than multi-state continuous progression dynamics.
11. **External Geographic Validation Nuance:** SEER is made up of several US geographic cancer registries with geographically diverse epidemiological coverage, but the models were not evaluated across international healthcare systems or different European registries. We see this as the most important next step for clinical translation and as deliberately outside the scope of the current methodological benchmark, whose principal contribution is the auditing template itself rather than a single deployable model. For a subsequent external-validation study, we consider METABRIC (Molecular Taxonomy of Breast Cancer International Consortium) an appropriate candidate for

direct follow-up, given its shared clinical staging variables and different European origin relative to the US-based SEER registry used in this study. Such validation would not constitute a revision of the current benchmark, but an independent empirical contribution in its own right and should be designed and reported as such.

12. **SHAP Subsampling for SEER:** Due to the computational cost of Kernel SHAP, attribution analysis for SEER (Figure 4c) was performed on a fixed random subsample of 100 of the 491 holdout patients, rather than the full holdout used for WBCD and Coimbra.
13. **Small-Sample Holdout Instability for Coimbra:** The Coimbra holdout lockbox contains only $n = 18$ patients, and so point estimates in Table 4 (e.g., 61.11% accuracy, 95% CI: 38.6%–79.7%) are associated with large sampling uncertainty; a single fixed-seed partition of this size is not, per se, a statistically stable estimate of generalization performance. Its conformal coverage was likewise only 83.33% (15/18), compared with the 95% nominal objective and 95.70% pooled development-set coverage; this discrepancy may reflect finite-sample variation and does not by itself establish systematic undercoverage. We report the lockbox as a single pre-registered one-time evaluation following our sealed-lockbox protocol (Section 2.3), but note that the pooled cross-validated estimate in Table 3 ($71.35\% \pm 8.70\%$, $n = 50$ folds) is the more statistically reliable summary of Coimbra performance. We suggest that future replications on similarly small cohorts report the holdout alongside bootstrap-resampled performance and coverage distributions rather than single point estimates.
14. **Gradient Boosting Baselines with Minimal Hyperparameter Search:** The grids for XGBoost and LightGBM—`n_estimators` $\in \{50, 100\}$, `max_depth` $\in \{3, 5\}$, and `learning_rate` $\in \{0.01, 0.1\}$, yielding eight combinations per library—were deliberately narrow so that nested cross-validation remained computationally feasible over 50 outer folds for WBCD and Coimbra and 35 outer folds for SEER. The failure of these tuned baselines to statistically significantly outperform the Unified pipeline (Table 3) should not be interpreted as evidence that gradient boosting cannot match or surpass Unified performance under a more exhaustive search; it is a statement about the tuning budget used in this benchmark. Future comparisons should evaluate broader, computationally matched search spaces, ideally using nested randomized or Bayesian optimization.

5 Conclusion

Here, we present TRUST-BC, a multicohort benchmark that demonstrates the cohort-dependent predictive cost of removing parametric and covariance data leakage. The Unified pipeline without leakage obtained 97.08% accuracy on WBCD, 71.35% accuracy on Coimbra, and AUROC = 0.732 on SEER. Formal statistical equivalence with the leaky baseline was established only for WBCD. Equivalence was not established for Coimbra and SEER within the pre-specified margins (accuracy and AUROC gaps of 0.086 and 0.014, respectively). Thus, TOST results represent cohort-specific effect sizes and uncertainty in their estimation, rather than established equivalence. This cohort-dependent pattern is a key finding in its own right: leakage effects are not to be uniformly assumed to be small across clinical tabular domains. TRUST-BC further incorporates split conformal prediction (approximately 95% empirical coverage) to flag ambiguous cases for expert review, alongside multidimensional probability calibration and Decision Curve Analysis. An independent Weibull AFT survival branch further attained a C -index of 0.722 and significant prognostic risk separation (log-rank $p = 1.02 \times 10^{-9}$) on the isolated SEER test set. Together, these safeguards separate true clinical signal from artifacts of flawed evaluation, and provide a transparent, reproducible benchmark for oncology machine learning researchers, consistent with TRIPOD+AI and PROBAST+AI principles. Further work should validate the framework against geographically distinct external registries and include wider audits of near-duplicate, site-specific, and temporal leakage.

Declarations

Ethics Approval and Consent to Participate: Ethical review was not required for this study as all analyses were conducted on publicly available, fully de-identified retrospective research datasets.

Consent for Publication: Not applicable.

Data and Code Availability: Complete source code, dataset manifests, and evaluation logs are publicly available at github.com/academic-repo/trust-bc-multicohort-benchmark and permanently archived

on Zenodo (DOI: 10.5281/zenodo.placeholder). WBCD and Coimbra datasets are accessible via the UCI Machine Learning Repository [12]. The curated SEER research benchmark extraction is accessible via public oncology benchmark archives [13].

Software and Computational Environment: Analyses were implemented in Python 3.10 utilizing scikit-learn (v1.6.1) [8], XGBoost (v3.4.1) [2], LightGBM (v4.6.0) [3], SHAP (v0.52.0) [33], NumPy (v2.1.3), SciPy (v1.16.3), and lifelines. This benchmark did not log machine-level telemetry, including CPU model, RAM, thread count, peak memory, and wall-clock runtime; these fields therefore cannot be reconstructed reliably for the reported runs. The next public reproducibility release will instrument the pipeline to capture this telemetry by default for all future executions. Users reproducing this template should consult the latest versioned release in the repository and retain its environment manifest and telemetry logs alongside the analytical outputs.

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of Interest: The authors declare that they have no competing financial interests or personal relationships that could have influenced this work.

TRIPOD+AI & PROBAST+AI Compliance: Item checklists are provided in Supplementary Material [34, 35].

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